

Ahmad Ghalib Athariq

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Introduction

Programmer with interest in systems-level software and modern AI/ML. I enjoy (self)-learning across domains and working on problems that demand both technical depth and curiosity.

Education

Universitas Sumatera Utara Medan, Indonesia
BS in Computer Science (S.Kom.) Sep 2021 – July 2025

- **Thesis Topic:** Computer Vision in Agriculture
- **Coursework Highlights:** Computer Organization & Architecture, Algorithms, Computer Vision, Computer Networks
- **Advisors:** Dr. Pauzi Ibrahim Nainggolan; Seniman S.Kom., M.Kom.
- **GPA:** 3.67/4.0 (distinction)

Work Experience

PT Qansa Solusi Indonesia (software company) Medan, Indonesia
Software Engineer Jan 2026 – Present

- Developing DiGasss (Digital Access and Asset Sumatra), a core enterprise platform powering field service and asset management across TelkomGroup's infrastructure operations.
- Built and maintained application features using **Laravel and Livewire**, ensuring user friendly UI/UX experience.
- Completed over **60+ high-priority tickets** over the span of three months.

Systatum (startup) Remote
Technical Intern Jan 2026 – July 2026

- Part of the Backend/DevOps Team responsible for maintaining and developing efficient, scalable systems.
- Leveraged **test-driven development** (TDD) principles to build a backend web application in Crystal (Ruby-inspired compiled language) using the Marten web framework.
- Improved development workflows by contributing **build automation scripts** and operational utility tools.

PT Xtend Integrasi Indonesia (software house) Medan, Indonesia
Backend Developer Nov 2024 – Jan 2025

- Developed **secure web APIs** for a TNI AD's information system and an in-house ERP product.
- Collaborated closely with the team lead to ensure correct delivery of requested features.

PT Len Industri (Persero) (defense technology company) Bandung, Indonesia
Desktop Developer Intern (Magang Merdeka) Feb 2024 – Jun 2024

- Contributed to the development of a **C2 (Command & Control)** application by successfully integrating it with the company's proprietary Data Link Processor software within **4 months**.
- Developed software components designed by the lead software engineer.
- Developed **comprehensive unit tests** for new and existing modules.
- Enhanced software stability and reliability by meticulously **analyzing large output logs** to identify and resolve critical bugs.

Revolutionary Games Studio (game development team) Remote (international)
Volunteer Programmer Jan 2020 – 2024

- Spearheaded development of gameplay features, user interface and contributed bug fixes.
- Contributed over **350+ commits** and **400k line additions** to the codebase – accelerating the development of the game in **3-year span**.
- Ensured codebase quality and maintainability by performing **rigorous code reviews** on pull request submissions to the project's GitHub repository.

Projects

Portfolio: me.alibata.web.id/projects (for additional information and older projects)

AI/LLM workflows builder (2025) 🔗 Repository Link

- Developed a **visual workflow builder** for creating and executing AI agent workflows as directed acyclic graphs with asynchronous execution via Redis queues.
- Leveraged **Google Gemini APIs** for agent/node capability in the workflow.
- Utilized Server-Sent Events (SSE) to provide real-time workflow execution status to the client.

- Tools Used: TypeScript, Express.js, React.js, TanStack Router, React Flow

Automatic lettuce-plant phenotype estimator (2025)

 [Repository Link](#)

- Developed a **lightweight object detector** (multi-task single-stage) that performs both detection and automatic per-plant fresh-weight (FW) estimation from RGB-D images.
- Achieved detection and estimation **accuracy of over 88% and 97%**, which is on-par or higher than a public benchmark.
- Runs **on-device deep learning model** using ExecuTorch: **216 ms/inference** (RGB-D variant) on a mobile-tier CPU.
- Tools Used: Python, PyTorch, OpenCV, ExecuTorch, Android

upahkita.id: indonesian salary data platform (active 2024 - Jun 2025)

[upahkita.id](#)

- Conceptualized and self-built a full-stack data platform to provide transparent, geographically-visualized salary data for the Indonesian market. **Was deployed to a \$3/mo VPS.**
- Designed a data ingestion pipeline with change data capture (CDC) to support full-text data search using MeiliSearch.
- Made to prove that building a full-blown end-to-end web app is possible alone and efficiently, by using today's tech stacks.
- Tools Used: Next.js, TypeScript, Golang, PostgreSQL, Meilisearch

as2roid: an interactive cmd-line game (2023)

 [Repository Link](#)

- Developed a Windows-native game inspired by space shooter arcade games, fully running on a **command-prompt window**.
- Integrated DirectSound to create a custom audio engine.
- Tools Used: Windows API (WinAPI), C++ , CMake, Visual Studio

Organization

Ikatan Mahasiswa Ilmu Komputer USU, Communications and Information Staff

Sep 2023 – Sep 2024

- Initiated a **design overhaul** of IMILKOM USU's official website: [imilkom.usu.ac.id](#)
- **Skills**: Trello, Teamwork, Communication, Responsibility, Patience

Google Developer Students Club (GDSC) USU, Member

2023 – 2024

Skills

Languages: Indonesian - Native, English - Bilingual

Programming Languages: C#, C++14, Python, Java, Golang, (Type/Java)Script, HTML, CSS, other webdev things

Technologies: Ubuntu, Windows, WSL, Android, Jupyter/Colab, Docker, PostgreSQL, VPS, Nginx

Developer Tools: VSCode, Visual Studio, Git, JetBrains IDEs, MSVC, CMake, vcpkg

Game Dev: Unity3d, Godot, Blender, Audacity, SDL

Other: LaTeX, Krita, Badminton (amateur)

Certifications

Magang Bersertifikat Batch 6

Certified Internship, PT Len Industri (Persero), Received 12 July 2024

Magang Generasi Bertalenta BUMN

00062/MAGENTA/FHCI01/9/24, Grade A, Received 4 September 2024

Cloud Practitioner Essentials

Dicoding, 1RXY1627QPVM, Received 22 December 2023

Back-End Application Basics

Dicoding, 07Z6WGKR2ZQR, Received 15 December 2023

Seminars

Fasilkom-TI USU

Participant, IEEE Educational Activities Committee Workshop on GenAI, 15 September 2025

Google Developer Group on Campus Indonesia

Participant, Talk Series 5: Integrating Embedded Systems for Autonomous Robotics Development, 26 August 2025

Volunteering

REFACTOR 2023

CS USU, Welcoming Freshmen, Media & Creative Team, September 2023

Student's General Elections Committee

USU, Student Executive Body, Media & Creative Team, December 2022

Computer Science Anniversary 2021

CS USU, Department Birthday Event, Media & Creative Team, December 2021

Miscellaneous Achievements

- Won 3rd place (9-person team) for best project in MKWK Fair USU 2022
- Top 5 in several local small-scale type racing competitions (2014 - 2024)
- 1st place on the leaderboard for plane dogfighting in Heroes & Generals (peak +10k online players, now defunct)